

Buildings of Generalized Braid Groups

Pierre Deligne (Bures-sur-Yvette)

Introduction

The main result of this work is the following.

Theorem. Let V be a finite-dimensional real vector space, let $\mathcal{M}$ be a finite set of homogeneous hyperplanes of V , let V_C be the complexification of V , and let

$$Y = V_C - \bigcup_{M \in \mathcal{M}} M_C$$

Assume that the connected components of $V - \bigcup_{M \in \mathcal{M}} M$ are open simplicial cones. Then Y is a $K(\pi, 1)$.

Let V be as above, and let $W \subset \text{GL}(V)$ be a finite group generated by reflections. Suppose that no nonzero vector of V is fixed by W :

$$V^W = 0$$

Let Φ be any Euclidean structure on V invariant under W , and let $\mathcal{M}$ be the set of hyperplanes M such that the orthogonal reflection with respect to M belongs to W . It is then known that $(V, \mathcal{M})$ satisfies the hypothesis of the theorem, and that W acts freely on the corresponding space Y_W . The quotient $X_W = Y_W/W$ is therefore also a $K(\pi, 1)$.

This result had been conjectured by Brieskorn. It is new only for W of type H_3 , H_4 , E_6 , E_7 , or E_8 (Brieskorn [2]). We also give a new proof that the fundamental group of X_W is the generalized braid group $\widetilde{W}$ corresponding to W (Brieskorn [3]).

Reduced to the particular case considered above, the plan of the proof is as follows.

- a) We construct a building $I(\widetilde{W})$ on which $\widetilde{W}$ acts, and a set $\mathcal{S}$ of spheres in $I(\widetilde{W})$, isomorphic to the unit sphere of V . The group $\widetilde{W}$ acts strictly simply transitively on $\mathcal{S}$.
- b) We show that, up to homotopy, $I(\widetilde{W})$ is the wedge of the spheres $S \in \mathcal{S}$.
- c) We relate X_W and $I(\widetilde{W})$.

The description of $\mathcal{S}$ and the possibility of c) had emerged during a conversation with Brieskorn and Tits in the spring of 1970. The ideas required to establish b) were supplied to me by Garside [4], whom I follow very closely in many places.

In Section 4, we determine the center of $\widetilde{W}$ and solve the word problem and the conjugacy problem in $\widetilde{W}$. These results were obtained independently by E. Brieskorn and K. Saito, who, like us, use Garside's methods [4].

The geometric language used and the techniques of proof are essentially due to Tits; I was able to familiarize myself with his ideas by attending one of his seminars and through conversations.

I am pleased to express my gratitude to him here.

0. Notation

(0.1) $L^+(D)$: the free monoid with identity generated by a set D .

(0.2) $L(D)$: the free group generated by a set D .

(0.3) A^- : the closure of a subset A of a topological space.

(0.4) For x, y in a monoid with identity and $n \geq 0$, set

$$\text{prod}(n; x, y) = xyxy \dots \quad (n \text{ factors});$$

one has $\text{prod}(2n; x, y) = (xy)^n$, $\text{prod}(2n+1; x, y) = (xy)^n x$.

1. Galleries

(1.1) Let $\mathcal{M}$ be a finite set of hyperplanes (the walls) in a finite-dimensional real vector space V . We shall use the terminology (walls, chambers, faces, facets) of [1] V § 1 (the facets form a partition of V). The support P of a facet F is the intersection of the walls containing F ; F is open in P . For every chamber A and every wall M , let $D_M(A)$ denote the set of chambers on the same side of M as A . If A and B are two chambers, let $D(A, B)$ denote the intersection of the $D_M(A)$ for which A and B are on the same side of M . Departing from the terminology of [1] IV 1 ex 15, we shall say that two chambers A and B are adjacent if $A \neq B$ and A and B have a face in common. We shall use the following elementary facts throughout.

(1.2) Lemma. (i) Let M be a wall of a chamber B . There exists a unique chamber B' adjacent to B and having M as a wall. M is the only wall separating B from B' .

(ii) Let B_1, B_2, B_3 be three chambers, and let $\mathcal{M}(B_i, B_j)$ be the set of walls separating B_i from B_j . Then

$$\mathcal{M}(B_1, B_3) = (\mathcal{M}(B_1, B_2) - \mathcal{M}(B_2, B_3)) \cup (\mathcal{M}(B_2, B_3) - \mathcal{M}(B_1, B_2))$$

A gallery of length n ($n \geq 0$), with source A and target B , is a sequence of chambers $(C_0, \dots, C_n)$ with $A = C_0$, C_{i+1} adjacent to C_i ($0 \leq i \leq n$), and $C_n = B$. The composite of two galleries $G = (C_0, \dots, C_n)$ and $G' = (C'_0, \dots, C'_m)$

is defined if $C_n = C'_0$, and is then

$$GG' = (C_0, \dots, C_n, C'_1, \dots, C'_m).$$

If G is a gallery with source A , $A.G$ denotes its target.

The opposite gallery G^* of a gallery $G = (C_0, \dots, C_n)$ is the gallery $(C_n, \dots, C_0)$. One has $(GG')^* = G'^* G^*$.

The antipodal gallery $-G$ of a gallery $G = (C_0, \dots, C_n)$ is the sequence of antipodal chambers $-G = (-C_0, \dots, -C_n)$.

One has $-(GG') = (-G)(-G')$, and $-(G^*) = (-G)^*$.

The distance $d(A, B)$ from a chamber A to a chamber B is the least length of a gallery from A to B . A gallery from A to B is minimal if it has length $d(A, B)$.

(1.3) Proposition. The distance $d(A, B)$ is the number of walls separating A from B . For a gallery G from A to B to be minimal, it is necessary and sufficient that it cross once the walls separating A from B and zero times the others.

It is clear that every gallery from A to B crosses every wall separating A from B , and it is enough to prove the existence of a gallery G from A to B whose length equals the number k of walls separating A from B . We proceed by induction on k . The intersection of the $D_M(A)$, for M a wall of A , is reduced to A . Thus either $A = B$, in which case we take $G = (A)$, or there is a wall M of A separating A from B . Let A' be the chamber adjacent to A having M as a wall. M is the only wall separating A from A' , so a wall N separates A' from B if and only if $M \neq N$ and N separates A from B . By the induction hypothesis, there is a gallery G' of length $k - 1$ from A' to B , and we take $G = (AA')G'$.

(1.4) Corollary. Let A, B, C be three chambers. The following conditions are equivalent.

(i) $d(A, C) = d(A, B) + d(B, C)$, i.e. there exists a minimal gallery from A to C passing through B .

(ii) For a wall to separate A from C , it is necessary and sufficient that it separate A from B or B from C . In other words, $\mathcal{M}(A, C) \subset \mathcal{M}(A, B) \cup \mathcal{M}(B, C)$ (cf. 1.2(ii)).

(iii) $B \in D(A, C)$.

(1.5) Proposition. Let P be an intersection of walls, let $V_P = V/P$, let $pr_P : V \rightarrow V_P$ be the projection, and let $\mathcal{M}_P$ be the set of hyperplanes N of V_P such that $pr_P^{-1}(N)$ is a wall. Denote by π'_P the unique map from the facets of $(V, \mathcal{M})$ to those of $(V_P, \mathcal{M}_P)$ such that $pr_P(F) \subset \pi'_P(F)$.

(i) π'_P respects the incidence relation $F_1 \subset F_2^-$: if $F_1 \subset F_2^-$, then

$$\text{codim}(F_1 \text{ in } F_2^-) \geq \text{codim}(\pi'_P[F_1] \text{ in } \pi'_P[F_2]).$$

π'_P takes chambers to chambers; if A and B are adjacent, either $\pi'_P(A) = \pi'_P(B)$, or $\pi'_P(A)$ and $\pi'_P(B)$ are adjacent.

(ii) Let F be a facet with support P (1.1). The restriction of π'_P to the set of facets E of $(V, \mathcal{M})$ such that $F \subset E^-$ is bijective. Like its inverse π_F , it preserves codimension, the incidence $E_1 \subset E_2^-$, and hence adjacency of chambers. We shall also write π_F for the bijection pr_P^{-1} between intersections of walls in V_P and intersections of walls in V containing P .

(iii) If $C = \pi_F(C')$, then, for every chamber X of $(V, \mathcal{M})$,

$$\pi_F^{-1}D(C, X) = D(C', \pi'_P(X)) \text{ and } \pi_F^{-1}\mathcal{M}(C, X) = \mathcal{M}_P(C', \pi'_P(X)).$$

For $X = \pi_F(X')$, one has

$\pi_F(D(C', X')) = D(C, X)$ and $\pi_F(\mathcal{M}_P(C', X')) = \mathcal{M}(C, X)$;

π_F then induces a bijection between minimal galleries from C' to X' and from C to X .

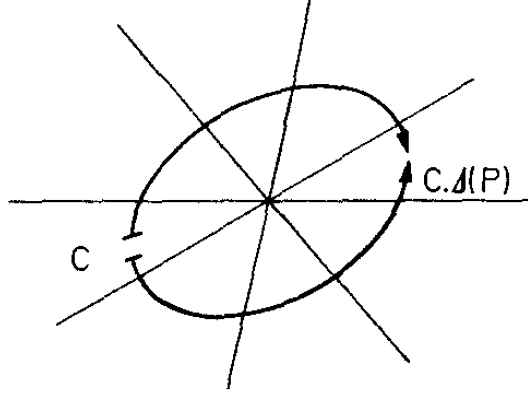
The verification is left to the reader ((iii) follows from 1.4(ii) $\iff$ (iii)).

- (1.6) Notation.** (i) For F a facet with support P , V_P , $\mathcal{M}_P$, pr_P , π'_P are also denoted V_F , $\mathcal{M}_F$, pr_F , π'_F .
(ii) Let P be an intersection of walls and C a chamber. Suppose that there is a facet of C (i.e. in C^-) with support P . This facet is then unique. Denote it by $F(P)$, and set

$$C.\Delta(P) = \pi_{F(P)}(-\pi_{F(P)}^{-1}(C))$$

- (1.7) Lemma.** (i) A wall separates C from $C.\Delta(P)$ if and only if it contains P ((1.5)(iii)).
(ii) If M is a wall of C , then $C.\Delta(M)$ is the unique chamber adjacent to C and having M as a wall (cf. 1.5(iii), 1.2(i)).
(iii) Let $M \neq N$ be two walls of C whose intersection P contains a facet of C open in P . There are exactly two minimal galleries from C to $C.\Delta(P)$. One begins with $(C, C.\Delta(M))$; the other begins with $(C, C.\Delta(N))$.
Assertion (iii) is verified by reduction to rank two (cf. 1.5(iii)), where the drawing is
(1.7.1)

(1.7.1)



From now on we assume the following condition.

- (1.8) Hypothesis.** The chambers are open simplicial cones.

In other words, each chamber is the set of points with coordinates > 0 in a suitable basis of V .

- (1.9.1) Remark.** Let P be an intersection of walls.

(i) $(V_P, \mathcal{M}_P)$ still satisfies (1.8);

(ii) the set $\mathcal{M}^P$ of traces on P of the walls not containing P still satisfies (1.8).

If $\mathcal{M}$ is defined by a “Weyl group” W (see the introduction), $\mathcal{M}^P$ is in general no longer of this type.

- (1.9.2) Remark.** Hypothesis (1.8) ensures that if P is an intersection of walls of a chamber C , the condition of (1.6) is satisfied. The chamber $C.\Delta(P)$ is therefore defined.

(1.10) Definition. Two galleries G and G' with the same endpoints are equivalent (notation: $G \sim G'$) if there exists a sequence of galleries $G = G_0, \dots, G_n = G'$ ($n \geq 0$) such that G_{j+1} is obtained from G_j in the following way ($0 \leq j < n$):

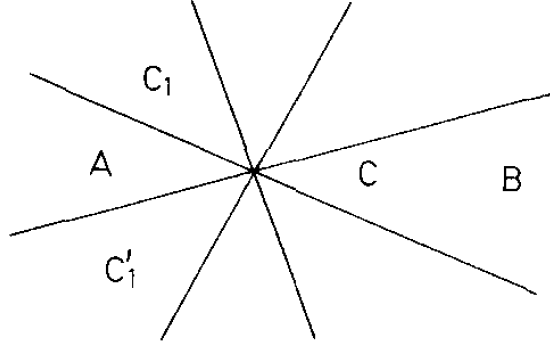
- a) there are decompositions $G_j = E_1 F E_2$, $G_{j+1} = E_1 F' E_2$;
b) there is a chamber C and two walls M, N of C such that F and F' are the two minimal galleries from C to $C.\Delta(M \cap N)$.

Composition of galleries, the operations $G \longrightarrow G^*$ and $G \longrightarrow -G$, the source and target maps, and the “length” function are compatible with this equivalence relation and therefore pass to the quotient. We shall generally denote classes of galleries by lowercase letters. We say that a class of galleries e begins (respectively ends) with a class of galleries f if there exists g such that $e = fg$ (respectively $e = gf$). We verify:

(1.11) Proposition. Equivalent galleries cross each wall the same number of times.

(1.12) Proposition. Two minimal galleries with the same endpoints are equivalent.

Let $G = (C_0, \dots, C_n)$ and $G' = (C'_0, \dots, C'_n)$, with endpoints A and B . We proceed by induction on the length of the galleries under consideration. The case $n = 0$ is trivial, and the induction hypothesis allows us to suppose that $C_1 \neq C'_1$. Let M and M' be the walls separating $A = C_0 = C'_0$ from C_1 and C'_1 , and let $C = A.\Delta(M \cap M')$. The walls M and M' separate A from B . By reduction to rank two (1.5)(iii), it follows that every wall separating A from C separates A from B .



(dessin dans $V_{M \cap M'}$; on omet d'écrire $\pi'_{M \cap M'}()$).

(drawing in $V_{M \cap M'}$; we omit writing $\pi'_{M \cap M'}()$).

Let F (respectively F') be the minimal gallery from A to C that passes through C_1 (respectively C'_1), and let E be a minimal gallery from C to B . The galleries FE and $F'E$ are minimal and equivalent. The minimal galleries G and FE (respectively G' and $F'E$) begin with (A, C_1) (respectively with (A, C'_1)). The induction hypothesis implies that they are equivalent, and (1.12) follows.

(1.13) Notation. (i) Let $u(A, B)$ denote the equivalence class of minimal galleries from A to B .

(ii) For I a set of walls of a chamber C , with intersection P , set $\Delta(P) = u(C, C.\Delta(P))$ (cf. (1.9.2)). For I the set of all walls, set $\Delta = u(C, -C)$.

The interest of the notation lies in its ambiguity with respect to C .

(1.14) Proposition. Let A be a chamber and $\mathcal{G}$ a set of classes of galleries of bounded length with source A . Suppose that $\mathcal{G}$ satisfies the conjunction (i) of

(i_a) $(A) \in \mathcal{G}$.

(i_b) If $gh \in \mathcal{G}$, then $g \in \mathcal{G}$.

(i_c) Let g have target B , and let M, N be two walls of B . If $g\Delta(M)$ and $g\Delta(N)$ are in $\mathcal{G}$, then $g\Delta(M \cap N) \in \mathcal{G}$.

Then

(ii) There exists a unique class of galleries x with source A such that

$\mathcal{G} = \{g \mid x \text{ begins with } g\}$.

Uniqueness is clear: if x begins with y and $x \neq y$, then y has strictly smaller length than x . Let x have maximal length in $\mathcal{G}$. To ensure that x has the required property, it is enough to prove the following assertion.

(*) Let g and M be such that (a) x begins with g and (b) x does not begin with $g\Delta(M)$, while $g\Delta(M) \in \mathcal{G}$. Then there exist g' and M' still satisfying (a) and (b), with g' strictly longer than g .

Since $g\Delta(M) \in \mathcal{G}$, maximality of x implies that $g \neq x$, and hence that x begins with $g\Delta(N)$ for a suitable N , necessarily distinct from M . By (i_c), one then has $g\Delta(M \cap N) \in \mathcal{G}$.

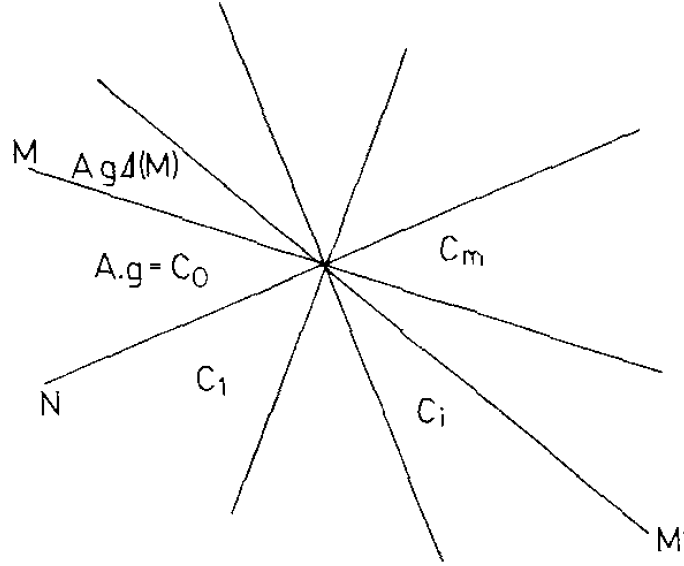
Since $g\Delta(M \cap N)$ begins with $g\Delta(M)$, x does not begin with $g\Delta(M \cap N)$.

Let $(C_0, \dots, C_m)$ be the minimal gallery from $A.g$ to $A.g\Delta(M \cap N)$ that begins with $(A.g, A.g\Delta(N))$. Let i ($0 < i < m$) be the largest i such that x begins with $g' = g(C_0, \dots, C_i)$. Then g' and the wall M' between C_i and C_{i+1} satisfy (a)(b).

(projection in $V_{M \cap N}$).

(1.15) Proposition. Let A be a chamber and $\mathcal{B}$ a set of chambers. For $\mathcal{B}$ to satisfy the conjunction (i) consisting of

(i_a) $A \in \mathcal{B}$.



(projection dans $V_{M \cap N}$).

(i_b) If $B \in \mathcal{B}$, then $D(A, B) \subset \mathcal{B}$.

(i_c) Let M and N be two walls of a chamber B . If A and B are on the same side of M and of N , and $B.\Delta(M)$ and $B.\Delta(N)$ belong to $\mathcal{B}$, then $B.\Delta(M \cap N) \in \mathcal{B}$.

it is necessary and sufficient that

(ii) There exists a unique chamber C such that $\mathcal{B} = D(A, C)$.

It is easy to verify that (ii) $\Rightarrow$ (i). To prove that (i) $\Rightarrow$ (ii), apply (1.14) to the set $\mathcal{G}$ of the $u(A, B)$ for $B \in \mathcal{B}$ (note that, by (1.3) and (1.11), every gallery G of class g such that some $u(A, B)$ begins with g is automatically minimal), and apply (1.4)(i) $\iff$ (iii) to the gallery class $x = u(A, X)$ thereby obtained.

Applying criterion (1.15), we find:

(1.16) Corollary. Let A and B be two adjacent chambers separated by the wall M , and let C be on the same side of M as B . There exists C' such that

$$D(A, C) \cap D_M(B) = D(B, C')$$

(1.17) Corollary. (i) Let A, C_1, C_2 be three chambers. There exists C such that

$$(1.17.1) \quad D(A, C) = D(A, C_1) \cap D(A, C_2).$$

(ii) Let A and C_1 be two chambers, and let M be a wall of A . There exists C' such that

$$(1.17.2) \quad D(A, C') = D(A, C_1) \cap D_M(A).$$

In fact, (ii) is the special case of (i) with $C_2 = -(A.\Delta(M))$.

(1.18) Lemma. Let M, M', M'' be three distinct walls of a chamber A , $A_1 = A.\Delta(M)$, $B = A.\Delta(M \cap M' \cap M'')$, $B' = A.\Delta(M \cap M')$, and $B'' = A.\Delta(M \cap M'')$. For every chamber C , if $B' \in D(A_1, C)$ and $B'' \in D(A_1, C)$, then $B \in D(A_1, C)$.

The facet F of A open in $M \cap M' \cap M''$ is a facet of all the chambers A, A_1, B, B' , and B'' . Applying (1.5)(iii), we may therefore reduce to the case $\dim V = 3$. Applying (1.17)(ii), we may further suppose that A_1 and C are on the same side of M . Make these assumptions, and suppose that $B', B'' \in D(A_1, C)$. Since M' (respectively M'') separates A_1 from B' (respectively B''), A_1 and C are separated by M' and M'' (and are on the same side of M); hence A and C are separated by M, M' , and M'' , and $C = -A = B$.

The following key result is inspired by [4].

(1.19) Proposition. (i) Let A, B, C be three chambers, F a gallery from A to B , and G_1, G_2 two galleries from B to C . If $FG_1 \sim FG_2$, then $G_1 \sim G_2$.

(ii) Similarly, if $G_1 E \sim G_2 E$, then $G_1 \sim G_2$.

(iii) Let g be an equivalence class of galleries with source A . There exists a chamber C such that g begins with $u(A, B)$ if and only if $B \in D(A, C)$.

Statement (ii) follows from (i) by passing to opposite galleries.

Let $G = (C_0, \dots, C_n)$ and $G' = (C'_0, \dots, C'_n)$ be two galleries of length n with the same endpoints. If $n \geq 1$, set $G_1 = (C_1, \dots, C_n)$ and $G'_1 = (C'_1, \dots, C'_n)$. If moreover $C_1 \neq C'_1$, let M and M' be the walls separating $C_0 = C'_0$ from C_1 and C'_1 , and set $A = C_0 \Delta (M \cap M')$. Consider the following relation $R_n(G, G')$ between galleries G and G' as above.

$R_n(G, G') \iff$ one of the following holds:

$\alpha)$ $n = 0$;

$\beta)$ $n \neq 0$, $C_1 = C'_1$, and $G_1 \sim G'_1$;

$\gamma)$ $n \neq 0$, $C_1 \neq C'_1$, and there exists a gallery F from A to $C_n = C'_n$ such that $G_1 \sim u(C_1, A).F$ and $G'_1 \sim u(C'_1, A).F$.

We shall prove by induction on n the following assertion:

(A_n) R_n is an equivalence relation.

Assume (A_i) for $i < n$, and prove (1.19.1) through (1.19.3) below.

(1.19.1) For galleries of length $i < n$, $R_i(G, G') \iff G \sim G'$.

It is trivial that $R_i(G, G') \Rightarrow G \sim G'$; and, with the notation of (1.10), if $G \sim G'$, then $R_i(G_j, G_{j+1})$.

(1.19.2) Assertion (i) is true for FG_1 of length $< n$.

This follows from (1.19.1) and the second clause in the definition of the R_i .

(1.19.3) Assertion (iii) is true for g of length $< n$.

This follows from (1.15) applied to the set $\mathcal{B}$ of chambers B such that g begins with $u(A, B)$: hypothesis (i_c) of (1.15) is verified using (1.19.2), (1.18), and the third clause in the definition of the R_i .

It remains to prove that if G, G' , and G'' are three galleries of length n from A to C , and $R_n(G, G')$ and $R_n(G, G'')$, then $R_n(G', G'')$. If $n = 0$, or $C_1 = C'_1$, or $C_1 = C''_1$, this is trivial.

For $n > 0$, denote by M, M', M'' the walls separating $A = C_0 = C'_0 = C''_0$ from C_1, C'_1 , and C''_1 . Distinguish two cases.

Case 1. $C_1 \neq C'_1 = C''_1$. Let $B = A \Delta (M \cap M')$. By hypothesis,

$(C_1, \dots, C_n) \sim u(C_1, B).F' \sim u(C_1, B).F''$,

$(C'_1, \dots, C'_n) \sim u(C'_1, B).F'$,

$(C''_1, \dots, C''_n) \sim u(C''_1, B).F''$.

It follows from (1.19.2) that $F' \sim F''$, and hence that $R_n(G', G'')$.

Case 2. $C_1 \neq C'_1 \neq C''_1 \neq C_1$. Let $B_1 = A \Delta (M' \cap M'')$, $B' = A \Delta (M \cap M')$, $B'' = A \Delta (M \cap M'')$, and $B = A \Delta (M \cap M' \cap M'')$. We have

$$G_1 = (C_1, \dots, C_n) \sim u(C_1, B').F' \sim u(C_1, B'').F''$$

$$G'_1 = (C'_1, \dots, C'_n) \sim u(C'_1, B').F'$$

$$G''_1 = (C''_1, \dots, C''_n) \sim u(C''_1, B'').F''$$

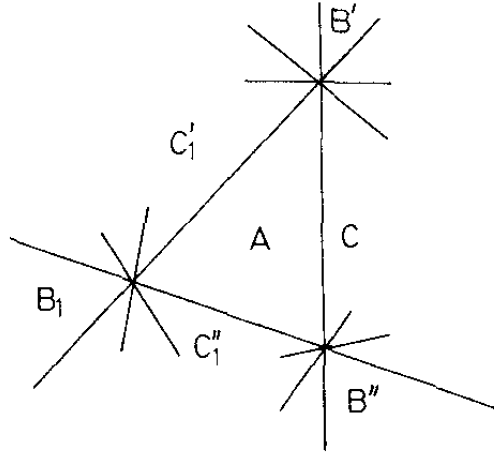
so that the class g_1 of G_1 begins with $u(C_1, B')$ and $u(C_1, B'')$. It then follows from (1.19.3) and Lemma (1.18) that g_1 begins with $u(C_1, B)$: $G_1 \sim u(C_1, B).F$. By (1.19.2), $F' \sim u(B', B).F$ and $F'' \sim u(B'', B).F$; hence $G_1 \sim u(C_1, B).F$, $G'_1 \sim u(C'_1, B).F$, and $G''_1 \sim u(C''_1, B).F$. Therefore,

$G'_1 \sim u(C'_1, B_1)u(B_1, B).F$

$G''_1 \sim u(C''_1, B_1)u(B_1, B).F$,

so that $R_n(G', G'')$.

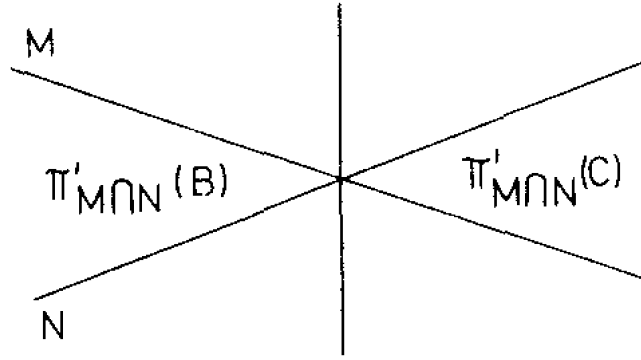
(drawing on the unit sphere of $V_{M \cap M' \cap M''}$).



(dessin sur la sphère unité de $V_{M \cap M' \cap M''}$).

(1.20) Corollary. Conditions (i) and (ii) of (1.14) are equivalent.

Assume (ii) and prove (i_c) . Under the hypotheses of (i_c) , if $x = gh$ (h with source B), then (1.19)(i) implies that h begins with $\Delta(M)$ and $\Delta(N)$. Let C be the chamber guaranteed by (1.19)(iii) for h . Since M and N separate B from C , $\pi'_{M \cap N}(C)$ can only be $\pi'_{M \cap N}(B) \cdot \Delta$, and (i_c) follows from (1.5)(iii).



(1.21) Corollary. For every chamber A , let $n(A, i)$ be the number of classes of galleries with source A and length i . Set

$$f_A = \sum_0^\infty n(A, i) t^i \in \mathbb{Z}[[t]]$$

For every chamber A , and for P running through the intersections of walls of A (including $\{0\}$ and V), one has

$$\sum_P (-1)^{\text{codim}(P)} t^{d(A, A \cdot \Delta(P))} f_{A \cdot \Delta(P)} = 1$$

This expresses that

- a) the number of classes of galleries of length i that begin with $\Delta(P)$ is $n(A \cdot \Delta(P), i - d(A, A \cdot \Delta(P)))$ ((1.19)(i));
- b) classes of galleries that begin with $\Delta(P)$ and $\Delta(Q)$ begin with $\Delta(P \cap Q)$ (by (1.19)(iii)).

(1.22) Algorithm. Let $G = (A_0, \dots, A_n)$ be a gallery of length $n \geq 1$, let $G_1 = (A_1, \dots, A_n)$, let M be the wall separating A_0 from A_1 , let C be the chamber whose existence is guaranteed by (1.19)(iii) (for G), and let C_1 be the analogous chamber for G_1 . One can compute C recursively using the following formula (cf. (1.16)):

$$(1.22.1) \quad D(A_1, C) = D_M(A_1) \cap D(A_1, C_1).$$

Since $A_1 \in D(A, C)$, M indeed separates A from C and $C \in D_M(A_1)$. By (1.19)(i), one also has $D(A_1, C) \subset D(A_1, C_1)$, giving the inclusion $\subset$. Finally, if $B \in D_M(A_1) \cap D(A_1, C_1)$, then G_1 begins with $u(A_1, B)$ and G begins with $u(A, A_1) \cdot u(A_1, B) = u(A, B)$, which proves (1.22.1).

(1.23) Corollary. Let G and H be two composable galleries. Let A be the source of G , B that of H , and let C be guaranteed by (1.19)(iii): one has $H \sim u(B, C).H'$. Then, for every chamber D , for the class of GH to begin with $u(A, D)$, it is necessary (and sufficient) that the class of $Gu(B, C)$ begin with $u(A, D)$.

(1.24) Proposition. Let A be a chamber, P an intersection of walls of A , $F = F(P)$ (1.6), and g an equivalence class of galleries with source A . There exists a class of galleries g' of $(V_P, \mathcal{M}_P)$, with source $\pi_F^{-1}(A)$, such that, for every gallery H of $(V_P, \mathcal{M}_P)$, g begins with $\pi_F(H)$ if and only if g' begins with H .

Let $\mathcal{G}$ be the set of classes of galleries h with source $\pi_F^{-1}(A)$ in $(V_P, \mathcal{M}_P)$ such that g begins with $\pi_F(h)$. Apply (1.14) to $(V_P, \mathcal{M}_P)$ and to $\mathcal{G}$. By (1.19), condition (i) is satisfied (cf. the proof of (1.20)), and (1.24) follows from (1.14)(ii).

(1.25) One may regard the set of galleries as the set of arrows of a category $\text{Gal}_0(V, \mathcal{M})$ whose objects are the chambers. Similarly, equivalence classes of galleries are the arrows of a quotient category $\text{Gal}_+(V, \mathcal{M})$. Let A, B , and C be three chambers, E a gallery from A to B , and F a gallery from B to C . Although the general categorical conventions are different, we shall continue to write EF for the composite of E and F . The operation $*$ (respectively $G \rightarrow -G$) is an anti-equivalence (respectively an equivalence) of $\text{Gal}_0(V, \mathcal{M})$ or $\text{Gal}_+(V, \mathcal{M})$ with itself, inducing the identity (respectively $C \rightarrow -C$) on the set of objects.

When there is no risk of confusion, we shall write simply Gal_0 and Gal_+ for $\text{Gal}_0(V, \mathcal{M})$ and $\text{Gal}_+(V, \mathcal{M})$ (or for a category $\text{Gal}_0(V_P, \mathcal{M}_P)$ or $\text{Gal}_+(V_P, \mathcal{M}_P)$). By abuse of language, we shall often call the arrows of Gal_+ galleries.

(1.26) Lemma. (i) In Gal_+ , for every gallery g , one has

$$g\Delta = \Delta(-g)$$

(ii) For any g and h with source A in Gal_+ , there exists n such that $g\Delta^n$ begins with h . If h is composed of k galleries $u(A_i, A_{i+1})$, one may take $n = k$.

Proceeding by induction, it is enough to prove (i) for g of length one. For $g = (B, C)$, one has

$$g\Delta = u(B, C)u(C, -C) = u(B, C)u(C, -B)u(-B, -C) = u(B, -B)u(-B, -C) = \Delta(-g)$$

For every chamber B , $\Delta = u(A, -A)$ begins with $u(A, B)$. Hence (ii) follows from (i) by induction on k .

The following proposition follows at once from (1.26), from its transform by $*$, and from (1.19)(i), (ii).

(1.27) Proposition. (i) In Gal_+ , the set of all arrows admits a left and right calculus of fractions.

(ii) Let $\text{Gal}(V, \mathcal{M})$, or simply Gal , be the category (a groupoid) obtained from Gal_+ by making all arrows invertible: call positive the arrows of Gal lying in the image of Gal_+ . The canonical functor from Gal_+ to Gal is faithful, and every arrow g of Gal can be written in the forms $g = g_1\Delta^{-n} = \Delta^{-n}g_2$ with g_1 and g_2 positive.

If a category C has only one object A and the monoid $\text{Hom}(A, A)$ is cancellative, saying that all arrows of C admit a left and right calculus of fractions means that $\text{Hom}(A, A)$ satisfies the Öre condition on the left and on the right. The “theory of the calculus of fractions” used in (1.27) and below is an immediate generalization of the theory of Öre for embedding such a monoid in a group.

(1.28) For every wall M , the number of times that g in Gal_+ crosses M is well defined (1.11). This function of g extends additively to $g \in \text{Gal}$. More generally, for every intersection of walls P , one could use the universal property of Gal to define functors

$$\pi'_P : \text{Gal}(V, \mathcal{M}) \longrightarrow \text{Gal}(V_P, \mathcal{M}_P)$$

(1.29) Let A be a chamber, P an intersection of walls of A , and $F = F(P)$ (1.6). The function π_F induces a functor

$$(1.29.1) \quad \pi_F : \text{Gal}_0(V_P, \mathcal{M}_P) \longrightarrow \text{Gal}_0(V, \mathcal{M}).$$

The image of this functor is stable under equivalence, and it induces a faithful functor

$$(1.29.2) \quad \pi_F : \text{Gal}_+(V_P, \mathcal{M}_P) \longrightarrow \text{Gal}_+(V, \mathcal{M}).$$

From the theory of the calculus of fractions and (1.19)(i) follows the

(1.30) Proposition. Under the preceding hypotheses, the functor deduced from (1.29.2)

$$\pi_F : \text{Gal}(V_P, \mathcal{M}_P) \longrightarrow \text{Gal}(V, \mathcal{M})$$

is faithful.

(1.31) Proposition. Let C be a chamber, I and J two sets of walls of C , $K = I \cap J$, P , Q , and R the intersections of the walls in I , J , and K , and $F(P)$, $F(Q)$, and $F(R)$ the corresponding facets in C (1.6). The facet $F(R)$ is the smallest facet containing $F(P)$ and $F(Q)$. In the groupoid $\text{Gal}(V, \mathcal{M})$, one has

$$\pi_{F(P)}(\text{Gal}(V_P, \mathcal{M}_P)) \cap \pi_{F(Q)}(\text{Gal}(V_Q, \mathcal{M}_Q)) = \pi_{F(R)}(\text{Gal}(V_R, \mathcal{M}_R))$$

A chamber having $F(P)$ and $F(Q)$ as facets also has $F(R)$ as a facet. This proves (1.31) for the objects.

Let $g \in \text{Hom}_{\text{Gal}}(A, B)$, and suppose that

$$g = \pi_{F(P)}(g'_1) = \pi_{F(Q)}(g'_2)$$

By (1.27)(ii), one has

$$g'_1 = \Delta(P)^{-n} g''_1 \text{ and } g'_2 = \Delta(Q)^{-n} g''_2$$

with g''_i positive, for $n \geq 0$ sufficiently large. Let $g_1 = \pi_{F(P)} g''_1$ and $g_2 = \pi_{F(Q)} g''_2$.

By (1.26)(ii) (transformed by $*$), $\Delta^n \Delta(P)^{-n}$ is positive; in $\text{Gal}_+(V, \mathcal{M})$ one has

$$(1.31.1) \quad (\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2.$$

(1.31.2) Lemma. If $h \in \text{Hom}_{\text{Gal}_+}(A, B)$ begins both with $(\Delta^n \Delta(P)^{-n})$ and with $(\Delta^n \Delta(Q)^{-n})$, then h begins with $(\Delta^n \Delta(R)^{-n})$.

Let us deduce (1.31) from (1.31.2). By (1.31.2), one has

$$(\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2 = (\Delta^n \Delta(R)^{-n}) g_3$$

with g_3 positive. Thus $g = \Delta(R)^{-n} g_3$. The positive gallery $g_3 = \Delta(R)^n g$ belongs to the image of $\pi_{F(P)}$ and of $\pi_{F(Q)}$. By (1.28), it therefore crosses zero times every wall not containing P or Q , i.e. not containing R . It consequently belongs to the image of $\pi_{F(R)}$.

Let us prove (1.31.2). Set $A(P) = A \cdot \Delta \cdot \Delta(P)$, and similarly for Q and R . The gallery $\Delta \Delta(P)^{-1}$ with source A is $u(A, A(P))$; the one with source $A(P)$ is $u(A(P), A)$. It follows from (1.26)(i) that $\Delta \Delta(P) = \Delta(P) \Delta$. Hence

$$(1.31.3) \quad \Delta^n \Delta(P)^{-n} = u(A, A(P)) \cdot u(A(P), A) \cdot u(A, A(P)) \dots \quad (n \text{ factors}).$$

$$(1.31.4) \text{ Lemma. } A(R) \in D(A(P), A(Q)).$$

With the notation of (1.2), $\mathcal{M}(A \Delta, A(P))$ is the set of walls containing P , and $\mathcal{M}(A \Delta, A(Q))$ the set of those containing Q . By (1.2), $\mathcal{M}(A(P), A(Q))$ is the set of those containing P or Q , but not R , namely $\mathcal{M}(A(P), A(R)) \cup \mathcal{M}(A(Q), A(R))$; apply (1.4).

By (1.19)(iii), a gallery class that begins with $\Delta \Delta(P)^{-1}$ and $\Delta \Delta(Q)^{-1}$ therefore also begins with $\Delta \Delta(R)^{-1}$, and (1.31.2) follows by induction from

(1.31.5) Lemma. Let P and R be intersections of walls of a chamber A , with $P \subset R$. Let h be in Gal_+ with source A . If h begins both with $\Delta^n \Delta(P)^{-n}$ ($n > 0$) and with $\Delta \Delta(R)^{-1}$, then h begins with $(\Delta \Delta(R)^{-1})(\Delta \Delta(P)^{-1})^{n-1}$.

We may suppose that $n \geq 2$. Set $h = (\Delta \Delta(P)^{-1}) h'$. With the preceding notation, h' then begins both with $\Delta \Delta(P)^{-1} = u(A(P), A)$ and with $u(A(P), A(R))$. Apply (1.19)(iii).

The chamber C of loc. cit. is then separated from $A(P)$ by every wall M that separates $A(P)$ from A (i.e. $M \not\supset P$) or $A(P)$ from $A(R)$ (i.e. $M \supset P$ and $M \not\supset R$). Thus C is separated from $A(P) \Delta$ only by walls $M \supset R$, and $C \in D(A(P) \Delta, A(P) \Delta \Delta(R))$. Hence h' begins both with $(\Delta \Delta(R)^{-1})$ and with $(\Delta \Delta(P)^{-1})^{n-1}$. Proceeding by induction on n , one deduces that h' begins with $(\Delta \Delta(R)^{-1})(\Delta \Delta(P)^{-1})^{n-2}$, so that h begins with $(\Delta \Delta(P)^{-1})(\Delta \Delta(R)^{-1})(\Delta \Delta(P)^{-1})^{n-2}$; the conclusion follows by noting that

$$\Delta \Delta(P)^{-1} \Delta \Delta(R)^{-1} = \Delta \Delta(R)^{-1} \Delta \Delta(P)^{-1}$$

(1.32) Proposition. For F a facet of a chamber C , spanning an intersection of walls P , one has

$$\pi_F^{-1}(\text{Gal}_+(V, \mathcal{M})) = \text{Gal}_+(V_P, \mathcal{M}_P) \subset \text{Gal}(V_P, \mathcal{M}_P)$$

Suppose that $\pi_F(\Delta^{-n}g) = h$ lies in $\text{Gal}_+(V, \mathcal{M})$. Then $\pi_F(g) = \Delta(P)^n h$ and, by (1.28), every gallery H representing h crosses only walls passing through F . Thus $h = \pi_F(h_1)$, with h_1 in $\text{Gal}_+(V_P, \mathcal{M}_P)$, and the conclusion follows from (1.30).

2. Buildings

(2.1) Let $(V, \mathcal{M})$ be as in § 1 (satisfying (1.8)), and let $r = \dim V$. Let S be the sphere of radius one in V , for any Euclidean structure on V (more intrinsically, one could set $S = V - \{0\}/\mathbb{R}_+^*$). The hyperplanes $M \in \mathcal{M}$ cut out a triangulation of S , and we shall also denote by S the corresponding simplicial space (simplicial scheme) and its geometric realization. We transfer to S the terminology used for V (chambers, facets, adjacency, gallery, ...).

(2.2) Choose a “fundamental chamber” A_0 in S . We denote by I_+ the space constructed below; it depends on $V, \mathcal{M}$, and A_0 , and is equipped with $q : I_+ \longrightarrow S$.

a) Let $\mathcal{L}_+$ be the set of equivalence classes of galleries g with source A_0 ,

$$\mathcal{L}_+ = \coprod_B \text{Hom}_{\text{Gal}_+}(A_0, B)$$

b) Let Z be the disjoint sum, indexed by $g \in \mathcal{L}_+$, of the closed simplex of S that is the closure of the open simplex of dimension $r - 1$ which is the target of g :

$$Z_+ = \coprod_{g \in \mathcal{L}_+} (\text{target of } g)^-.$$

Let q' be the evident map from Z_+ onto S .

c) I_+ (equipped with q) is a quotient of Z_+ (equipped with q'). If G is a gallery from A_0 to B and C is a chamber adjacent to B , one glues $(\text{target of } g)^-$ and $(\text{target of } g(BC))^-$ along the common closed face of their images in S .

(2.3) The space I_+ is decomposed into facets, images of facets of Z_+ , each mapping bijectively onto a facet of S . Its chambers (respectively

faces and vertices) are its facets of dimension $r - 1$ (respectively $r - 2$ and 0).

The vertices of a facet F are the vertices contained in the closure $\overline{F}$ of F . The chambers of I_+ are indexed by $\mathcal{L}_+$; denote by $\tilde{A}_0.g$ the one with index g and by $\tilde{A}_0$ the one with index (A_0) .

Let $g \in \text{Hom}_{\text{Gal}_+}(A_0, A)$, let $\tilde{A} = \tilde{A}_0.g$, and let $H = (C_0, \dots, C_n)$ be a gallery from A to B , of class h .

Set

$$\tilde{A}.h = \tilde{A}_0.gh$$

The chambers $\tilde{A}.(C_0, \dots, C_i)$ ($0 \leq i \leq n$) form a gallery in I_+ . Galleries obtained in this way will be called positive.

(2.4) Lemma. Let F be a facet of S , let A and B be two chambers of S having F as a facet, let $g \in \text{Hom}_{\text{Gal}_+}(A_0, A)$, and let $h = \text{Hom}_{\text{Gal}_+}(A_0, B)$. The following conditions are equivalent:

- (a) in I_+ , the facets $q^{-1}(F)$ of $\tilde{A}_0.g$ and $\tilde{A}_0.h$ coincide;
- (b) $g^{-1}h \in \text{Hom}_{\text{Gal}}(A, B)$ lies in $\pi_F(\text{Gal})$;
- (c) there exist g_1 and h_1 in $\pi_F(\text{Gal}_+)$ such that $gg_1 = hh_1$.

Condition (b) is an equivalence relation among galleries from A_0 to a chamber having F as a facet. It follows that (a) $\Rightarrow$ (b). That (b) $\Rightarrow$ (c) follows from (1.26)(i). Finally, (c) $\Rightarrow$ (a) is immediate from the definitions.

From (2.4)(a) $\iff$ (b) and (1.31), one immediately obtains the following result.

(2.5) Proposition. Each facet of I_+ is uniquely determined by the set of its vertices. Thus I_+ may be described as the geometric realization of the following simplicial scheme.

a) For x a vertex of S , let $\mathcal{G}(x)$ be the set of g in Gal_+ with source A_0 and target a chamber having x as a vertex. Let $\mathcal{G}(x)/\pi_x$ be the quotient of $\mathcal{G}(x)$ by the equivalence relation (2.4)(b) (for $F = x$). Then the set of vertices is

$$\coprod_x \mathcal{G}(x)/\pi_x$$

b) For a set E of vertices to span a simplex, it is necessary and sufficient that there exist a gallery g with source A_0 such that, for $y \in E$, g lies in $\mathcal{G}(qy)$ and y is the image of g in $\mathcal{G}(qy)/\pi_{qy}$.

(2.6) Proposition. Let F be a facet of I_+ . There exists a gallery class g such that, for F to be a facet of a chamber $B = \tilde{A}_0.h$, it is necessary and sufficient that

$$h = g\pi_F(h')$$

for a suitable h' in Gal_+ .

Take g to be a gallery of minimum length such that F is a facet of $A = \tilde{A}_0.g$. Let $B = \tilde{A}_0.h$ be a chamber having F as a facet. By (2.4)(a) $\iff$ (c), there exist galleries h_1 and h_2 in $(V_{qF}, \mathcal{M}_{qF})$ such that

$$g\pi_F(h_1) = h\pi_F(h_2)$$

Apply the transform by $*$ of (1.24). In view of the minimality of g , one finds that h_1 ends with h_2 ; $h_1 = h'h_2$. By (1.19)(ii), one has

$$g\pi_F(h') = h,$$

which proves (2.6).

(2.7) Notation. Let A be a chamber of I_+ . Denote by $S(A)$ the union of the closed chambers $(A.(qA, B))^-$, as B ranges over the chambers of S .

One verifies that $q|S(A)$ is an isomorphism between $S(A)$ and S .

(2.8) Definition. $\hat{I}_+$ is the space obtained from I_+ by “filling in” the spheres $S(A)$.

More precisely, let B^r be a ball whose boundary is S . When working simplicially, take B^r to be the cone on the simplicial space S . Define $\hat{I}_+$ from I_+ by attaching a family of copies $b(A)$ of B^r , indexed by the set of chambers of I_+ . The attaching maps are

$$\partial b(A) \xrightarrow{\sim} S \xleftarrow{q} S(A).$$

The attaching maps are simplicial, so $\hat{I}_+$ again appears as the geometric realization of a simplicial scheme (whose vertices are those of I_+ together with the “centers of the balls”). The space $\hat{I}_+$ is equipped with an evident (simplicial) map

$$q : \hat{I}_+ \longrightarrow B^r$$

(2.9) Proposition. The space $\hat{I}_+$ is contractible.

The space I_+ is therefore the wedge of the spheres $S(A)$.

Let $(I_+)_n$ be the union of the closed chambers $(\tilde{A}_0.g)^-$ for g of length $\leq n$, and let $(\hat{I}_+)_n$ be the union of $(I_+)_n$ and the set of balls $b(A)$ for which $\partial b(A) \subset (I_+)_n$.

One has $(\hat{I}_+)_0 = \tilde{A}_0$ (contractible), and

$$\hat{I}_+ = \varinjlim (\hat{I}_+)_n$$

The proposition follows from

(2.9.1) Lemma. $(\hat{I}_+)_n$ is a deformation retract of $(\hat{I}_+)_n$.

We must construct a continuous family $(\varphi_t)_{0 \leq t \leq 1}$ of continuous maps from $(\hat{I}_+)_{n+1}$ to itself, with $\varphi_0 = \text{Id}$, $\varphi_t|(\hat{I}_+)_n = \text{Id}$, and $\varphi_1((\hat{I}_+)_{n+1}) = (\hat{I}_+)_n$.

Let $A = \tilde{A}_0.g$ be a chamber of $(I_+)_{n+1}$ that is not in $(I_+)_n$, and let F be a facet of A .

(2.9.2) Lemma. Suppose that there exists a chamber $B = \tilde{A}_0.h$ in $(I_+)_{n+1}$, distinct from A , having F as a facet. Then there exists a chamber $B' = \tilde{A}_0.h'$ in $(I_+)_n$ and a wall M of qB' , containing qF , such that $A = B'.\Delta(M)$.

Let g_0 be the gallery considered in (2.6). One has

$$g = g_0\pi_F(g_1) \text{ and } h = g_0\pi_F(h_1).$$

The hypotheses imply that the length $\lg(g_1) \neq 0$ (otherwise, since $A \neq B$, one would have $\lg(h) > \lg(g) = n+1$). For a suitable wall M of qA , one then has $g_1 = h'\Delta(M)$, and set $B' = \tilde{A}_0.h'$.

For the closed chamber A^- , $A^- \cap (I_+)_n$ is therefore the union of a nonempty set of closed faces; and, in $(I_+)_{n+1}$, the points of $A^- - (A^- \cap (I_+)_n)$ belong only to the single closed chamber A^- .

We distinguish two cases.

Case 1. $A^- \cap (I_+)_n \neq \partial A^-$.

In this case, there is no sphere $S(B)$ with

$$A \subset S(B) \subset (I_+)_{n+1}.$$

We take $\varphi_t|A^-$ to be a collapse of A^- onto $A^- \cap (I_+)_n$.

Case 2. $A^- \cap (I_+)_n = \partial A^-$.

Set $A = \tilde{A}_0.g$. For every wall M of qA , it then follows from (2.7.2) that g ends with $\Delta(M)$.

By ((1.19)(iii)), g therefore ends with Δ : one has $A = B.\Delta$, with B uniquely determined by A , by (1.19)(ii). In passing from $(\hat{I}_+)_{n+1}$ to $(\hat{I}_+)_n$, A and the interior of the ball $b(B)$ disappear. We take $\varphi_t|b(B)$ to be a collapse of $b(B)$ onto $S(B) - A$.

By what was seen in case 1, every ball that disappears is of the preceding type. This completes the construction of φ_t and proves (2.7).

(2.10) The space I , equipped with $q : I \rightarrow S$, is defined like I_+ , replacing Gal_+ by Gal .

a) Set $\mathcal{L} = \coprod_B \text{Hom}_{\text{Gal}}(A_0, B)$.

b) Set $Z = \coprod_{g \in \mathcal{L}} (\text{target of } g)^-$.

c) Let q' be the evident map from Z onto S . The space I , equipped with q , is a quotient of Z equipped with q' . For $g \in \mathcal{L}$ with target B and C a chamber adjacent to B , one glues $(\text{target of } g)^-$ and $(\text{target of } g(BC))^-$ along the common closed face of their images in S . In other words, if two

chambers B and C have a common facet F , if $g \in \text{Hom}_{\text{Gal}}(A_0, B)$ and $h \in \text{Hom}_{\text{Gal}}(A_0, C)$, then in I the closed facets $q'^{-1}(F)^-$ of $(\text{target of } g)^-$ and of $(\text{target of } h)^-$ are identified if and only if

$$hg^{-1} \in \pi_F \text{Hom}_{\text{Gal}}(\pi_F^{-1}(B), \pi_F^{-1}(C)).$$

(2.11) As for I_+ , one defines the chambers, faces, and open or closed facets of I . One defines $\tilde{A}_0$ and $\tilde{A}.h$ (for $\tilde{A}$ a chamber and h in Gal with source $q\tilde{A}$) as in (2.3). The analogue of (2.4)(a) $\iff$ (b) is here trivially true. As in (2.5), one then deduces from (1.31) that each facet of I is uniquely determined by the set of its vertices, and I appears as the geometric realization of a simplicial scheme admitting a description of type (2.5). For every chamber A of I , define a sphere $S(A)$ as in (2.7); q induces an isomorphism from $S(A)$ onto S . As in (2.8), define a building $\hat{I}$, equipped with $q : \hat{I} \rightarrow B^r$, by “filling in” all the spheres $S(A)$.

(2.12) Besides q and its decomposition into chambers, I has the following additional structure:

— A “composition” $\tilde{B}.g$ is defined which, to a chamber $\tilde{B}$ of I with image B in S and to $g \in \text{Hom}_{\text{Gal}}(B, C)$, associates a chamber of I with image C in S . The construction $g \mapsto \tilde{B}.g$ defines an isomorphism from the space analogous to I obtained by taking B as fundamental chamber to the space I (for $A_0 = B$, one obtains automorphisms of I). In particular,

$$\tilde{B}.(gh) = (\tilde{B}.g).h.$$

(2.13) For every chamber A of I lying over A_0 , define a map

$$i_A : I_+ \rightarrow I \text{ (respectively } \hat{I}_+ \rightarrow \hat{I}),$$

compatible with the projection onto S (respectively B^r), by sending the closed chamber $(A.g)^-$ of I_+ (respectively the ball $b(A.g)$ of $\hat{I}_+$) to the chamber (respectively the ball) of the same name in I (respectively $\hat{I}$).

(2.14) Proposition. (i) i_A identifies I_+ with a closed subspace of I and $\hat{I}_+$ with a closed subspace of $\hat{I}$.

(ii) One has

$$I = \varinjlim_n i_{\tilde{A}_0.\Delta^{-2n}}(I_+) \text{ and } \hat{I} = \varinjlim_n i_{\tilde{A}_0.\Delta^{-2n}}(\hat{I}_+).$$

The assertions (i) or (ii) for $\hat{I}_+$ and $\hat{I}$ follow from the assertions (i) or (ii) for I_+ and I .

We prove (i) for I_+ and I . Let B and C be two chambers of S having a common facet F . Let $g \in \text{Hom}_{\text{Gal}_+}(A_0, B)$ and $h \in \text{Hom}_{\text{Gal}_+}(A, C)$. Suppose that the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are equal in I .

One then has

$g^{-1}h = \pi_F(e)$ for $e \in \text{Hom}_{\text{Gal}}(\pi_F^{-1}B, \pi_F^{-1}C)$.

By (1.26), one may write $e = e_1 e_2^{-1}$ with e_i in Gal_+ . By (1.29), one then has $g\pi_F(e_1) = h\pi_F(e_2)$ in Gal_+ , and the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are therefore already equal in I_+ .

To prove (ii), it suffices to observe that every gallery g with source A_0 in Gal can be written $g = \Delta^{-2n}g'$ with g' in Gal_+ .

From (2.14) one obtains $\pi_i(\widehat{I}) = \varinjlim \pi_i(\widehat{I}_+)$. Since I is a CW complex, (2.9) gives the following result.

(2.15) Theorem. The space $\widehat{I}$ is contractible.

The space I is therefore the wedge of the spheres $S(A)$, as A ranges over the chambers of I .

3. Coverings

(3.1) Let V_C be the complexification of V , and M_C the complexification of M for $M \in \mathcal{M}$, and let

$$Y = V_C - \bigcup_{M \in \mathcal{M}} M_C$$

In this section we construct, by gluing, a space $\widetilde{Y}$ above Y . The gluing data will be described using I and $\widehat{I}$. The facets of S and the facets of V other than $\{0\}$ are in canonical bijection, and we shall constantly pass from one to the other. In general, we use the same notation for corresponding facets; when necessary, the correspondence will be denoted $F \mapsto [F]$.

(3.2) Lemma. Let A , B , and C be three chambers of I , with $C \subset S(A) \cap S(B)$. Then q induces an isomorphism from $S(A) \cap S(B)$ onto the intersection of S with the closed half-spaces $D'_M(qC)$ containing qC and bounded by a wall M separating qA from qB .

The intersection of the $D'_M(qC)$ is a union of closed chambers. If C' is one of them, qA and qB lie on the same side of every wall separating qC from C' . It follows that the lifts to $S(A)$ and $S(B)$ of a minimal gallery from qC to C' coincide, and $(C')^- \subset q(S(A) \cap S(B))$.

Conversely, if a facet F is not in the intersection of the $D'_M(qC)$, there is a wall M such that $F \not\subset M$ which separates qA from qB and F from qC . Suppose, impossibly, that $F \subset q(S(A) \cap S(B))$. Let D be a chamber of S having F as a facet, and let $(C_0, C_1 \dots C_{n+1})$ be a gallery from qC to D . Let M_i be the wall separating C_i from C_{i+1} , and let α_i (respectively β_i) = ± 1 according as C_{i+1} and A (respectively B) are or are not on the same side of M_i . By hypothesis, F is a facet of $C\Delta(M_0)^{\alpha_0} \dots \Delta(M_n)^{\alpha_n}$ and of $C.\Delta(M_0)^{\beta_0} \dots \Delta(M_n)^{\beta_n}$.

By (2.4) and (1.24), one then has (since $M \not\supset F$)

$$\sum_{M_i=M} \alpha_i = \sum_{M_i=M} \beta_i,$$

which is absurd: one side is one, the other -1 .

(3.3) Lemma. Let $\mathcal{R}$ and $\mathcal{I}$ be the “real part” and “imaginary part” maps from V_C to V . Let $v \in V_C$, and let F be the facet of V such that $\mathcal{R}v \in F$. For v to belong to Y , it is necessary and sufficient that $pr_F(\mathcal{I}v)$ lie in a chamber of $(V_F, \mathcal{M}_F)$.

This is clear.

What follows has as its intuitive basis the fact that to every “walk” in I (made by following a gallery of the form $\Delta(M_1)^{\varepsilon_1} \dots \Delta(M_k)^{\varepsilon_k}$, $\varepsilon_i = \pm 1$) there corresponds a path ch (a path class) in Y . The image under $\mathcal{R}$ of ch in V passes from chamber to chamber, successively crossing the walls $M_1, \dots, M_k$ at points m_i . For $\varepsilon_i = 1$ (respectively -1), the path passes through one of the two components R of $\mathcal{R}^{-1}(m_i) \cap Y$ (in bijection under $\mathcal{I}$ with $V - M$): the one for which $\mathcal{I}R$ contains (respectively does not contain) the preceding chamber.

(3.4) Notations. (i) Let C be a chamber of $(V, \mathcal{M})$. Denote by $Y(C)$ the open subset of Y consisting of the $x \in V_C$ satisfying the following condition:

— if F is the facet of V such that $\mathcal{R}x \in F$, then $pr_F \mathcal{I}x \in \pi'_F(C)$.

(ii) Let A and B be two chambers of I . If $S(A) \cap S(B)$ contains a chamber C , temporarily denote by $Y'(A, B)$ the intersection of the open half-spaces $D'_M(qC)^\circ$, for M as in 3.2. Set

$$Y(A, B) = Y(qA) \cap Y(qB) \cap \mathcal{R}^{-1}Y'(A, B) \subset V_C.$$

If $S(A) \cap S(B)$ contains no chamber, set $Y(A, B) = \emptyset$.

(iii) For a facet F of V , its star $\text{Et}(F)$ is the union of the open facets G of $(V, \mathcal{M})$ such that $F \subset G^-$.

(iv) For a facet F of V and a chamber C of $(V_F, \mathcal{M}_F)$, set

$$V(F, C) = \{v \in V_C \mid \mathcal{R}v \in \text{Et}(F) \text{ and } pr_F \mathcal{I}v \in C\}.$$

(3.5) Lemma. Let G be a facet of V . The $V(F, C)$, for F a facet such that $G \subset F^-$ and C a chamber of V_F , form an open covering of $Y \cap \mathcal{R}^{-1}(\text{Et } G)$. In particular (take $G = \{0\}$), the $V(F, C)$ cover Y .

If $x \in Y \cap \mathcal{R}^{-1}(\text{Et } G)$ and $\mathcal{R}x \in F$, then x lies in one of the $V(F, C)$ (3.3).

(3.6) Let Y' be the disjoint sum, indexed by the balls $b(A)$ of $\widehat{I}$,

$$Y' = \coprod_{b(A)} Y(qA).$$

Denote by $Y'(b(A))$ the component indexed by $b(A)$. There is an evident projection $p' : Y' \rightarrow Y$. Let R be the following relation on Y' :

For $x \in Y'(b(A))$ and $y \in Y'(b(B))$, $R(x, y)$ means that $p'(x) = p'(y)$ and that $p'(x), p'(y) \in Y(A, B)$.

It is an equivalence relation because (cf. (3.2)) $Y(A, B) \cap Y(B, C) \subset Y(A, C)$. One has

(3.6.1) The R -saturation of an open subset of Y' is again open.

Set $\widetilde{Y} = Y'/R$. There is an evident projection

$$p : \widetilde{Y} \rightarrow Y$$

For every ball $b(A)$ of $\widehat{I}$, denote by $\widetilde{Y}(b(A))$ the image of $Y'(b(A))$ in $\widetilde{Y}$.

The following theorem implies the one with which the introduction begins.

(3.7) Theorem. $\widetilde{Y}$ is a contractible covering of Y .

A. $\widetilde{Y}$ is a covering of Y .

We shall show, for $V(F, C)$ as in (3.4)(iv), that $p^{-1}V(F, C)$ is a sum of copies of $V(F, C)$.

a) Let $b(A)$ be a ball of $\widehat{I}$. For B a chamber such that $F \subset B^-$, let the following be the chamber of I :

$$C'_B = A.u(A, B)u(\pi_F(C), B)^{-1}$$

By construction, $B \subset q(S(A) \cap S(C'_B))$, and in particular, if $F \neq \{0\}$,

(3.7.1) $[F] \subset q(S(A) \cap S(C'_B))$.

We prove that

(3.7.2) $\widetilde{Y}(b(A)) \cap p^{-1}V(F, C) \subset \bigcup_{F \subset B^-} \widetilde{Y}(b(C'_B))$.

Let x be in the left-hand side. Let G be the facet of V to which $\mathcal{R}px$ belongs. One has $F \subset G^-$. Choose B to be a chamber having G as a facet. By hypothesis, one has

$$pr_G \mathcal{I}px \in \pi'_G(A) \text{ and } pr_F \mathcal{I}px \in C.$$

Thus $\pi'_G(A) = \pi'_G(\pi_F C)$. It then follows from (3.2) that every chamber having G as a facet lies in $q(S(A) \cap S(C'_B))$, that px lies in $Y(A, C'_B)$, and hence that $x \in \widetilde{Y}(b(C'_B))$.

b) Set $Y'(F, C) = \bigcup_{q(C')=C} (Y'(b(C')) \cap p'^{-1}(V(F, C)))$. By a), set-theoretically,

$$p^{-1}V(F, C) = Y'(F, C)/R.$$

This is even an isomorphism of topological spaces, by (3.6.1), since $Y'(F, C)$ is open in Y' . By (3.2), the equivalence relation

induced by R on $\bigcup_{q(C')=C} Y'(b(C'))$ is trivial. We conclude by observing that, for every C' above C , $V(F, C) \subset Y(b(C'))$.

B. An open covering of $\widetilde{Y}$.

(3.7.3) Notations. (i) For a facet F of I ,

$$\widetilde{Y}(F) = \bigcup_{F \subset S(C)} (\widetilde{Y}(b(C)) \cap p^{-1}\mathcal{R}^{-1}\text{Et}(qF)).$$

(ii) For $o(A)$, the center of a ball $b(A)$ of $\widehat{I}$,

$$\widetilde{Y}(o(A)) = \widetilde{Y}(b(A)) \cap p^{-1}V(\{0\}, A).$$

(3.7.4) Lemma. (i) The $\widetilde{Y}(s)$, for s a vertex of $\widehat{I}$, form a covering of $\widetilde{Y}$ whose nerve is $\widehat{I}$ (i.e., a family of $\widetilde{Y}(s)$ has nonempty intersection if and only if the corresponding s are the vertices of a facet of $\widehat{I}$).

(ii) If a facet F of I has vertices $s_0, \dots, s_p$, then

$$\widetilde{Y}(F) = \widetilde{Y}(s_0) \cap \dots \cap \widetilde{Y}(s_p).$$

(iii) If $F \subset S(A)$, then $\tilde{Y}(o(A)) \cap \tilde{Y}(F)$ is contractible; $\tilde{Y}(o(A))$ is contractible.

(3.7.4.1) Let F_i ($i = 1, 2$) be two facets of I and C_i ($i = 1, 2$) two chambers of I such that $F_i \subset S(C_i)$ and $(\tilde{Y}(b(C_1)) \cap p^{-1}\mathcal{R}^{-1}\text{Et}(qF_1)) \cap (\tilde{Y}(b(C_2)) \cap p^{-1}\mathcal{R}^{-1}\text{Et}(qF_2)) \neq \emptyset$.

Then F_1 and F_2 span a facet F of I (unique by (2.5) – (2.11)), $F \subset S(C_i)$, and the preceding intersection coincides with

$$\tilde{Y}(b(C_1)) \cap \tilde{Y}(b(C_2)) \cap p^{-1}\mathcal{R}^{-1}\text{Et}(F).$$

By hypothesis,

$$Y(C_1, C_2) \cap \mathcal{R}^{-1}\text{Et}(qF_1) \cap \mathcal{R}^{-1}\text{Et}(qF_2) \neq \emptyset;$$

F_1 and F_2 lie in $S(C_1) \cap S(C_2)$ and span a facet F there. Moreover, $\text{Et}(qF_1) \cap \text{Et}(qF_2) = \text{Et}(qF)$, whence the stated formula.

From (3.7.4.1), one deduces that $\tilde{Y}(F_1) \cap \tilde{Y}(F_2)$ is empty if F_1 and F_2 do not span a common facet F , and otherwise equals

$$\bigcap_{F \subset S(C)} \tilde{Y}(bC) \cap p^{-1}\mathcal{R}^{-1}\text{Et}(F) = \tilde{Y}(F)$$

(take $C_1 = C_2$). This proves (ii) and part of (i).

(3.7.4.2) If $o(A) \neq o(B)$, then $\tilde{Y}(o(A)) \cap \tilde{Y}(o(B)) = \emptyset$.

If $qA = qB$, one has $\tilde{Y}(b(A)) \cap \tilde{Y}(b(B)) = \emptyset$. If $qA \neq qB$, one has

$$p\tilde{Y}(o(A)) \cap p\tilde{Y}(o(B)) = \emptyset.$$

(3.7.4.3) If $F \not\subset S(A)$, then $\tilde{Y}(o(A)) \cap \tilde{Y}(F) = \emptyset$.

If $F \subset S(B)$, then $F \not\subset S(A) \cap S(B)$, and

$$\text{Et}(qF) \cap Y(A, B) = \text{Et}(qF) \cap p(\tilde{Y}(b(A)) \cap \tilde{Y}(b(B))) = \emptyset.$$

It is clear that:

(3.7.4.4) If $F \subset S(A)$, then p induces an isomorphism from $\tilde{Y}(o(A)) \cap \tilde{Y}(F)$ onto $Y(qA) \cap \mathcal{R}^{-1}\text{Et}(qF)$.

Lemma (3.7.4) then follows from:

(3.7.5) Lemma. For every facet F of V and every chamber A ,

$$Y(A) \cap \mathcal{R}^{-1}\text{Et}(F)$$

is contractible.

Let v_0 be a vector such that $\mathcal{R}v_0 \in F$ and $\mathcal{I}v_0 \in A$. Then, for $0 \leq t \leq 1$, $\varphi_t : v \mapsto v_0 + t(v - v_0)$ is a map of $Y(A)$ and of $\mathcal{R}^{-1}\text{Et}(F)$ into themselves. One has $\varphi_0(Y(A) \cap \mathcal{R}^{-1}\text{Et}(F)) = \{v_0\}$ and $\varphi_1 = \text{Id}$.

C. Study of $\tilde{Y}(F)$ (F a facet of I).

By (3.5), (3.7.1), and (3.7.2), one has

(3.7.6) $\tilde{Y}(F) = \bigcup_{F \subset B^-} \tilde{Y}(b(B)) \cap p^{-1}\mathcal{R}^{-1}\text{Et}(F)$.

Choose a “fundamental chamber” $A_{F,0}$ in $(V_F, \mathcal{M}_F)$, and denote the corresponding building by I_F . Also choose $\tilde{A}_{F,0}^I$ in I above $\pi_F(A_{F,0})$. For every chamber $B = \tilde{A}_0.g_F$ of I_F , denote by $\pi_F(B)$ the chamber $\tilde{A}_{F,0}^I.\pi_F(g_F)$ of I . By (2.10), (3.7.6) may be rewritten

$$\tilde{Y}(F) = \bigcup_{B \text{ in } I_F} \tilde{Y}(b(\pi_F(B))) \cap p^{-1}\mathcal{R}^{-1}\text{Et}(F).$$

Let

$$Y_F = (V_F)_C - \bigcup_{M \in \mathcal{M}_F} M_C,$$

and let $\tilde{Y}_F$ be the covering of Y_F defined using I_F .

For any B and C in I_F , it follows from (1.30) that the chambers of S in $Y(\pi_F(B), \pi_F(C)) \cap \text{Et}(F)$ are exactly the $\pi_F(D)$, for D in $Y_F(B, C)$.

The gluings defining $\tilde{Y}(F)$ and $\tilde{Y}_F$ are therefore compatible, and there is a unique Cartesian diagram

(3.7.7)

$$(3.7.7) \quad \begin{array}{ccc} \tilde{Y}_F & \longleftarrow & \tilde{Y}(F) \\ \downarrow & & \downarrow \\ Y_F & \xleftarrow{\text{pr}_F} & \mathcal{R}^{-1} \text{Et}(F) \subset Y \end{array}$$

which sends $\tilde{Y}(b(\pi_F(B))) \cap p^{-1}\mathcal{R}^{-1}\text{Et}(F)$ into $\tilde{Y}_F(b(B))$ for B a chamber of I_F . The first vertical arrow therefore also the second are coverings.

It is easily verified that the map pr_F of (3.7.7) is a homotopy equivalence, whence

(3.7.8) Lemma. $\tilde{Y}_F$ and $\tilde{Y}(F)$ have the same homotopy type.

D. End of the proof.

We prove (3.7) by induction on $r = \dim V$. Let $\mathcal{U}$ be the open covering of $\tilde{Y}$ by the $\tilde{Y}(s)$, for s a vertex of $\hat{I}$. By (3.7.4)(i), $\hat{I}$ is the nerve of $\mathcal{U}$. By (3.7.4), (3.7.8), and the induction hypothesis, the nonempty intersections of open sets belonging to $\mathcal{U}$ are contractible. This implies [6] that $\tilde{Y}$ and $\hat{I} = \text{Nerf}(\mathcal{U})$ have the same homotopy type. The conclusion follows from (2.15).

4. Braid groups

(4.1) Let V be a finite-dimensional real vector space and $W \subset \text{GL}(V)$ a finite group generated by reflections. Assume that $V^W = \{0\}$. Let Φ be any Euclidean structure on V invariant under W , and let $\mathcal{M}$ be the set of hyperplanes M such that the orthogonal reflection with respect to M belongs to W . One knows that

- a) $(V, \mathcal{M})$ satisfies 1.6 ([1] V 3.9 prop. 7);
- b) W permutes the chambers of $(V, \mathcal{M})$ in a strictly simply transitive manner ([1] V 3.2 Th. 1);
- c) if $x \in V$ belongs to the closed chamber $\bar{A}$, its stabilizer is generated by the reflections in the walls of A containing x ([1] V 3.3 prop. 1);
- d) (follows from c)) the group W acts freely on

$$Y_W = V_C - \bigcup_{M \in \mathcal{M}} M_C.$$

By b), if C_0 and C_1 are two chambers, there is a unique $w \in W$ such that $wC_0 = C_1$; these w induce a transitive system of bijections between the walls of the different chambers. Passing to the quotient, one obtains a set D with $\dim(V)$ elements and, for each chamber C , a bijection φ_C from D onto the set of walls of C . One has

(4.1.1) $\varphi_{wC}(i) = w\varphi_C(i)$ ($w \in W$).

For the definition of the Coxeter matrix $(m_{ij})_{i,j \in D}$ and that of the Coxeter graph, we refer to [1] V 3.4.

(4.2) Choose a chamber A_0 of $(V, \mathcal{M})$. For $i \in D$, let w_i be the reflection in the wall $\varphi_{A_0}(i)$. [1] V 3.2 Th. 1 says that W is generated by the w_i , these generators being subject only to the relations

$$(w_i w_j)^{m_{ij}} = e.$$

(4.3) Let $X_W = Y_W/W$: by (4.1)d), Y_W is a covering of X_W , with group W . Let $y_0 \in A_0$ have image $x_0 \in X_W$. For each $i \in D$, let ℓ'_i be the homotopy class of paths in Y_W , from y_0 to $w_i(y_0)$, which, for $y \in A_0$, contains the broken line with successive vertices $y_0, y_0 + iy_0, w_i(y_0) + iy_0, w_i(y_0)$. The image ℓ_i of ℓ'_i in X_W is a loop based at x_0 .

(4.4) Theorem. (i) The fundamental group $\pi_1(X_W, x_0)$ is generated by the ℓ_i ; these are subject only to the relations

(4.4.1) $\text{prod}(m_{ij}; \ell_i, \ell_j) = \text{prod}(m_{ji}; \ell_j, \ell_i)$.

(ii) The universal covering of X_W is contractible: X_W is a $K(\pi; 1)$.

Assertion (i) is proved in Brieskorn [3]. Another proof will be sketched in (4.18) below. As explained in the introduction, (ii) follows from (3.7) and (4.1).

(4.5) In § 1, we have drawn heavily on Garside's study [4] of Artin's braid group; from the results obtained we can now derive information about all groups of the type $\pi_1(X_W, x_0)$ (Garside had already considered some of them). The translation rests on (4.6), (4.7) below.

(4.6) Let $G = (C_0, \dots, C_n)$ be a gallery and M_i the wall separating C_{i-1} from C_i ($1 \leq i \leq n$). One has $\varphi_{C_{i-1}}^{-1}(M_i) = \varphi_{C_i}^{-1}(M_i)$. Let $\ell(G)$ be the element $\varphi_{C_1}^{-1}(M_1) \dots \varphi_{C_n}^{-1}(M_n)$ of $L^+(D)$ (0.1).

The map ℓ induces a bijection from the set of galleries G with a given source to $L^+(D)$. One has

(4.6.1) $\ell(w(G)) = \ell(G)$ (for $w \in W$);

(4.6.2) $\ell(G_0 G_1) = \ell(G_0) \ell(G_1)$ (for composable G_0 and G_1).

Let $w : L(D) \longrightarrow W$ be the homomorphism extending the map $i \longmapsto w_i$ from D to W .

(4.7) **Proposition.** For every gallery G with source A_0 , one has

$$A_0.G = w(\ell(G))(A_0).$$

This is an avatar of the identity

$$\begin{aligned} &[(w_1 \dots w_{n-1})w_n(w_1 \dots w_{n-1})^{-1}] \\ &\cdot [(w_1 \dots w_{n-2})w_{n-1}(w_1 \dots w_{n-2})^{-1}] \dots [w_1] = w_1 \dots w_n. \end{aligned}$$

(4.8) From this proposition one deduces that, for $w \in W$, the integer $d(A_0, w(A_0))$ is the least length of words in the w_i that equal w ; more precisely, ℓ induces a bijection from the set of minimal galleries from A_0 to wA_0 to the set of words $m \in L(D)$, of minimal length, such that $w = w(m)$.

(4.9) **Definition.** (i) The braid monoid G^+ of type D is the monoid with unit generated by D , the generators $i \in D$ being subject only to the relations

(4.9.1) $\text{prod}(m_{ij}; i, j) = \text{prod}(m_{ij}; j, i)$.

(ii) The braid group G (or generalized braid group) of type D is the group generated by D , the generators being subject to (4.9.1).

(4.10) Notice that W also admits the presentation

(4.10.1) $\text{prod}(m_{ij}; w_i, w_j) = \text{prod}(m_{ij}; w_j, w_i)$,

(4.10.2) $w_i^2 = e$.

so that $i \longmapsto w_i$ extends to a homomorphism from G to W .

(4.11) One checks from the definitions that two galleries G_0 and G_1 with the same source are equivalent if and only if $\ell(G_0)$ and $\ell(G_1)$ have the same image in G^+ . Taking account of (4.9), (1.12) therefore admits the following translation ([1] IV 1.6 prop. 5).

(4.12) **Reminder.** Let $w \in W$. The image in G^+ of the words $m \in L(D)$, of minimal length among those such that $w = w(m)$, depends only on w .

This image will be denoted $r(w)$; r is a section of $w : G^+ \longrightarrow W$. By (4.7), one has $r(w) = \ell(u(A_0, wA_0))$.

Similarly, [1] IV 1.4 lemma 2 corresponds to (1.11).

(4.13) The “quotient” category of $\text{Gal}_+(V, \mathcal{M})$ by W is defined as follows:

a) Its set of objects is reduced to one element: it is

$$\text{Ob}(\text{Gal}_+(V, \mathcal{M}))/W.$$

b) The monoid of its arrows is $\text{Fl}(\text{Gal}_+(V, \mathcal{M}))/W$, and passage to the quotient

$$\text{Gal}_+(V, \mathcal{M}) \longrightarrow \text{Gal}_+(V, \mathcal{M})/W$$

is a functor.

This category, having only one object, is identified with the monoid of its arrows. By (4.6.1), (4.6.2), and (4.11), ℓ identifies this monoid with G^+ .

Similarly one defines $\text{Gal}(V, \mathcal{M})/W$. This category has only one object and, by its universal property, ℓ extends to an isomorphism from the group of its arrows to G :

$$\begin{array}{ccccc} \text{Gal}_+(V, \mathcal{M}) & \longrightarrow & \text{Gal}_+(V, \mathcal{M})/W & \xrightarrow[\ell]{\sim} & G^+ \\ \downarrow & & \downarrow & & \downarrow \\ \text{Gal}(V, \mathcal{M}) & \longrightarrow & \text{Gal}(V, \mathcal{M})/W & \xrightarrow[\ell]{\sim} & G \end{array}$$

(4.14) Theorem. (i) In G^+ , left and right translations are injective.

(ii) G^+ satisfies the Öre condition on the left and on the right. It follows from (i) that G^+ embeds in G .

(iii) For J a subset of D , let G_J^+ (resp. G_J) be the submonoid (resp. subgroup) of G^+ (resp. G) generated by the $i \in J$. Then the evident map identifies G_J^+ (resp. G_J) with the braid monoid (resp. braid group) of type J .

(iv) One has $G_J^+ = G_J \cap G^+$.

(v) If J and K are two subsets of D , one has $G_J \cap G_K = G_{J \cap K}$.

(vi) Let $n(i)$ be the number of elements of G^+ of length i , and put $f = \sum n(i)t^i$. For $J \subset D$, let $m(J)$ be the number of walls passing through the intersection of the walls of an (arbitrary) chamber A , indexed by J (= the number of reflections in the Weyl group W_J). One has

$$f = \left(\sum_{J \subset D} (-1)^{|J|} t^{m(J)} \right)^{-1}$$

These assertions translate respectively: (i): (1.1)(i) and (ii); (ii): (1.26)(ii) and its transform by passage to opposite galleries (cf. also (4.16) below); (iii): (1.30); (iv): (1.32); (v): (1.31); (vi): (1.21) (the f_A of loc. cit. are all equal to f).

Let $x, y \in G^+$. We say that x begins with y if there exists z in G^+ with $x = yz$. Proposition (1.19)(iii) translates as follows

(4.15) Lemma. Let $x \in G^+$. There exists a chamber C such that, for every $w \in W$, x begins with $r(w)$ if and only if $wA_0 \in D(A_0, C)$.

(4.16) Let J be a subset of D and P the intersection of the walls $\varphi_{A_0}(i)$ ($i \in J$). Let $\Delta(J)$ be the element $\ell(u(A_0, A_0.\Delta(P)))$ of G^+ ; put $\Delta = \Delta(D)$. Let $i \mapsto \bar{i} = \varphi_{-C}^{-1} \varphi_C(i)$ be the opposition involution of D . We also denote by $w \mapsto \bar{w}$ the automorphisms of $L^+(D)$, G^+ , and G that send i to $\bar{i}$ (one has $m_{ij} = m_{\bar{i}\bar{j}}$). Translating (1.26) with the aid of (4.4), one finds that for g in G^+ or G ,

$$(4.16.1) \quad g\Delta = \Delta\bar{g}.$$

In particular, Δ^2 is central.

For every $i \in D$, Δ “begins” with i : one has $\Delta = ix$, in G^+ . Hence

(4.17) Proposition. G is obtained from G^+ by making the central element Δ^2 invertible.

(4.18) The description (2.11), (2.5) – (2.7) of the building I attached to $(V, \mathcal{M})$ can be translated as follows.

a) The set of vertices of I is

$$I^0 = \coprod_{i \in D} G/G_{D-i}$$

- b) For a set of vertices to span a simplex, it is necessary and sufficient that it be contained in the set of classes gG_{D-i} , for some $g \in G$.
- c) The fundamental sphere $S_0 = S(A_0)$ is the union of the fundamental simplex, spanned by the lateral classes of e in the G/G_{D-i} , and of its transforms by the $r(w)$ ($w \in W$).

The preceding description makes evident a left action of G on I . It respects the set of spheres $S(A)$ and the correspondence $A \mapsto S(A)$.

By transport of structure, G acts on the covering $\tilde{Y}_W$ of Y_W (3.6), and one has an equivariant morphism $(G - \tilde{Y}_W) \rightarrow (W - Y_W)$.

It follows that G is the fundamental group of $X_W = Y_W/W$, and (4.4)(i).

(4.19) For completeness, let us show that the word problem, the conjugacy problem, and the question of computing the center of G can be solved as in [4]. If the Coxeter graph D of W is a disjoint union of graphs D_α , the braid group of type D is the product of the braid groups of types D_α . The essential case is therefore that of a connected graph D .

(4.20) The word problem: there is a decision procedure for determining whether two elements m and n of $L(D)$ have the same image in G .

- a) Whether m and n in $L^+(D)$ have the same image in G^+ is decidable, because the imposed relations do not change the length of the words, and there are only finitely many words of a given length. One may also use (1.22).
- b) For any m_i ($i = 1, 2$) in the free group $L(D)$ generated by D , one can compute $m'_i \in L^+(D)$ and $k \geq 0$ such that m_i and $m'_i \Delta^{-k}$ have the same image in G (cf. (4.17)); m_1 and m_2 then have the same image in G if and only if m'_1 and m'_2 have the same image in G^+ .

(4.21) Theorem. If the Coxeter graph D is connected (nonempty) and the opposition involution is trivial (resp. nontrivial), the center of G is infinite monogenic, generated by Δ (resp. Δ^2).

Consider the following property of $x \in G^+$.

(*) For every $i \in D$, there exists $i' \in D$ such that $ix = xi'$.

It is enough to prove that, if x satisfies (*) and $x \neq e$, then $x = \Delta y$ with $y \in G^+$. By induction on the length of x , it will follow that if x satisfies (*), x is of the form Δ^k . In particular, the center of G^+ is contained in the set of the Δ^k ($k \geq 0$); by (4.14)(i) and (4.17), the center of G is contained in the set of the Δ^k ($k \in \mathbb{Z}$), and the assertion follows from (4.16.1).

Let $x \in G^+$ satisfy (*). Let C be as in (4.15), and let w be the image in W of $\ell u(A_0, C)$. We may write $x = r(w)y$, with $y \in G^+$. Let $i \in D$. By (*), $ix = r(w)yi'$ begins with $r(w)$; by (1.23), it follows that $ir(w)$ begins with $r(w)$: $ir(w) = r(w)i''$. In particular, for every i , there exists i'' with $iw = wi''$; this means that every wall of A_0 is also a wall of wA_0 . Among the $2^{|D|}$ open simplicial cones bounded by the walls of A_0 , only A_0 and $-A_0$ are chambers: they are the only ones whose angles between the faces are all $\leq \pi/2$ (the connectedness of D is used here). If $x \neq e$, then $w \neq e$ and hence $wA_0 = -A_0$. Since then $\Delta = r(w)$, this completes the proof.

(4.22) The derived group. Let D' be the graph having D as its set of vertices, two vertices being joined by an edge if and only if m_{ij} is odd. Let D_0 be the set of connected components of D' . Define an epimorphism

$$\text{Lg}: G \rightarrow \mathbb{Z}^{D_0}$$

by sending i to the basis vector indexed by the connected component of i in D' . One verifies at once that Lg identifies $\mathbb{Z}^{D_0}$ with the largest abelian quotient of G .

(4.23) The conjugacy problem. The problem is to give a decision procedure for determining whether two elements x and y of G (given explicitly as images of words in $L(D)$) are conjugate. Since Δ^2 is central, $\Delta^{-2k}a$ and $\Delta^{-2k}b$ are conjugate if and only if a and b are. By

(4.17), it therefore suffices to treat the case where $x, y \in G^+$. Similarly, if x and y are conjugate, there exists a in G^+ such that $xa = ay$. If x and y are conjugate, $\text{Lg}(x) = \text{Lg}(y)$. There are only finitely many $x \in G^+$ of given "length" $\text{Lg}(x)$, and W is finite. The existence of a decision procedure then follows from (4.20) and the following lemma.

(4.24) Lemma. Let x, y be in G^+ . If x and y are conjugate, there exists a sequence $x_0 = x, x_1, \dots, x_n = y$ of elements of G^+ and a sequence of elements w_i of W such that

$$x_{i+1}r(w_i) = r(w_i)x_i.$$

Let $a \in G^+$ be such that $xa = ay$. Put $a = r(w_0) \dots r(w_{n-1})$, where w_i is the element of W of maximum length such that $r(w_i) \dots r(w_{n-1})$ can be written in the form $r(w_i)b_i$ with $b_i \in G^+$ (4.15). Let $a_i = r(w_0) \dots r(w_i)$. We shall prove that $x_i = a_i^{-1}xa_i$ lies in G^+ , so that the x_i and w_i answer the problem.

It follows from (1.23) that, for every $w \in W$, if xa begins with $r(w)$, then $xr(w_0)$ also begins with $r(w)$. In particular, since $xa = ay = r(w_0)b_0y$ begins with $r(w_0)$, $xr(w_0) = r(w_0)x_1$ with $x_1 \in G^+$. The proof is completed by induction on n .

Remark. Let σ be an automorphism of the Coxeter graph D , and again denote by σ the corresponding automorphism of G . One has $\Delta^\sigma = \Delta$. The preceding arguments still apply to the question of deciding, for $x, y \in G$, whether there exists a in G such that $xa = a^\sigma y$. We reduce to taking x, y , and a in G^+ . In that case and, with the preceding notation, if $xa = a^\sigma y$, the $(a_i^{-1})^\sigma x a_i$ lie in G^+ . Taking for σ the involution $i \mapsto \bar{i}$, one obtains a criterion analogous to (4.24) for the conjugacy of Δx and Δy ($x, y \in G^+$).

Bibliography

The reader will find a more complete bibliography on braid groups in [2].

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Pierre Deligne
Institut des Hautes Études Scientifiques
35 Route de Chartres
F-91440 Bures-sur-Yvette
France

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